

BENGHAZI, Libyan Arab Jamahiriya

WMO: 620530

Lat: 32.097N

Lon: 20.269E

Elev: 132

StdP: 99.75

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.9	7.9	-3.8	2.8	23.6	-1.2	3.5	23.3	20.7	15.3	17.6	15.2	2.4	120	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.5	37.1	20.5	35.2	20.5	33.8	20.3	24.9	29.7	24.3	29.1	23.7	28.6	7.0	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	18.9	26.9	23.0	18.0	26.8	22.2	17.2	26.2	76.3	29.4	73.8	29.2	71.4	28.6	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.9	14.3	12.9	4.9	41.5	1.1	2.1	4.0	43.0	3.4	44.3	2.8	45.4	2.0	47.0		
			3.9	26.8	0.8	2.0	3.3	28.2	2.8	29.4	2.4	30.5	1.8	32.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.6	13.3	13.4	15.7	19.2	23.1	25.6	26.5	27.3	26.4	22.7	18.5	14.8
	DBStd	5.87	1.74	2.32	3.30	3.99	4.20	3.27	1.83	1.58	2.65	2.69	2.57	2.21
	HDD10.0	2	1	1	0	0	0	0	0	0	0	0	0	0
	HDD18.3	577	157	141	98	35	4	0	0	0	0	1	27	114
	CDD10.0	3858	102	96	176	276	407	467	511	535	493	394	254	148
	CDD18.3	1391	0	2	16	61	153	217	253	277	243	137	31	3
	CDH23.3	13494	1	10	131	580	1573	2271	2585	2934	2386	881	132	11
CDH26.7	5679	0	2	32	214	751	1073	1068	1251	1014	248	25	1	
Wind	WSAvg	5.8	5.3	5.6	5.9	6.8	6.6	6.4	6.4	6.1	5.5	5.1	5.1	5.3
Precipitation	PrecAvg	255	63	41	28	6	4	1	1	0	3	22	28	63
	PrecMax	369	212	134	112	39	25	7	26	5	17	79	67	161
	PrecMin	142	22	2	0	0	0	0	0	0	0	0	1	0
	PrecStd	73	35	31	24	8	6	2	5	1	4	22	21	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.7	24.9	30.1	34.8	39.0	40.8	37.2	37.0	38.1	34.1	30.0	24.8
		MCWB	11.9	13.8	15.4	17.9	21.7	20.9	21.4	21.6	21.5	20.3	16.3	13.7
	2%	DB	19.1	21.6	26.4	31.9	35.8	37.0	35.1	34.8	35.8	31.3	26.5	21.8
		MCWB	12.6	13.1	14.4	16.5	19.8	20.1	21.3	22.0	21.3	19.1	16.2	14.2
	5%	DB	18.0	19.3	24.0	29.1	33.2	34.4	33.0	33.1	33.4	29.2	24.8	20.0
		MCWB	12.5	12.6	13.8	15.9	18.2	19.7	21.4	22.3	20.8	19.0	16.2	14.0
	10%	DB	17.0	17.9	21.9	26.8	30.9	32.1	31.2	31.9	31.5	27.7	23.1	18.9
		MCWB	12.3	12.3	13.4	15.0	17.7	19.2	21.7	22.3	20.9	18.9	16.0	13.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.0	15.9	18.6	22.6	24.2	25.3	25.3	26.3	25.6	23.5	20.6	17.2
		MCDB	17.9	21.1	24.6	29.8	34.2	34.0	30.2	31.1	31.3	28.0	24.6	21.8
	2%	WB	14.0	14.5	16.4	18.7	21.5	23.1	24.3	25.0	24.3	21.9	18.8	15.6
		MCDB	17.3	19.2	22.2	27.4	30.6	29.2	28.8	29.3	29.6	26.7	22.8	19.2
	5%	WB	13.3	13.7	15.1	17.1	20.1	22.1	23.7	24.4	23.5	21.1	17.8	14.9
		MCDB	16.6	17.7	20.3	25.0	28.7	27.9	28.3	28.9	28.8	26.1	21.8	18.6
	10%	WB	12.7	13.0	14.3	16.0	19.0	21.4	23.2	23.9	22.7	20.3	16.9	14.1
		MCDB	16.0	16.5	19.7	23.4	26.4	27.6	27.8	28.5	28.0	25.2	21.1	17.7
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.0	9.7	10.9	11.0	10.8	9.5	9.5	9.9	9.0	8.6	7.3
		MCDBR	8.7	10.3	12.9	14.0	13.7	13.6	12.3	11.4	12.1	10.7	10.0	8.9
		MCWBR	5.2	5.5	6.0	6.4	6.1	6.2	5.6	5.2	5.5	4.9	5.1	5.1
	5% WB	MCDBR	7.7	9.0	10.8	12.4	12.0	11.0	9.8	9.5	9.5	9.0	8.7	8.0
		MCWBR	5.2	5.8	6.2	6.5	6.3	5.7	4.6	4.7	5.5	5.1	5.5	5.3
Clear-Sky Solar Irradiance	taub	0.368	0.400	0.449	0.498	0.486	0.435	0.426	0.429	0.456	0.424	0.389	0.374	
	taud	2.292	2.171	2.015	1.902	1.968	2.163	2.206	2.212	2.125	2.207	2.262	2.290	
	Ebn at Noon	838	844	828	802	814	853	859	853	811	809	806	807	
	Edh at Noon	107	133	167	195	184	151	144	141	148	127	110	103	
All-Sky Solar Radiation	RadAvg	3.11	3.91	5.43	6.52	7.29	7.97	7.98	7.26	6.01	4.57	3.42	2.75	
	RadStd	0.15	0.25	0.25	0.25	0.31	0.22	0.11	0.12	0.16	0.20	0.15	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page